

Week – 1 Finding Time Complexity of Algorithms

Q1. Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
    int s =1;
    while(s <= n)
    {
        i++;
        s += i;
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2
3 int count = 0;
4
5 void function(int n){
6     int i=1;
7     count++;
8     int s=1;
9     count++;
10    while(s<=n){
11        count++;
12        count++;
13        i++;
14        s+=i;
15        count++;
16    }
17    count++;
18    printf("%d",count);
19 }
20
21 int main(){
22     int n;
23     scanf("%d",&n);
24     function(n);
25 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Q2. Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2
3 int count = 0;
4 void func(int n){
5     if(n==1){
6         count++;
7         printf("*");
8     } else {
9         count++;
10        for(int i=1; i<=n; i++){
11            count++;
12            for(int j=1; j<=n; j++){
13                count++;
14                count++;
15                count++;
16                break;
17            }
18            count++;
19        }
20        count++;
21    }
22    printf("%d",count);
23 }
24
25 int main(){
26     int n;
27     scanf("%d",&n);
28     func(n);
29 }
```

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Passed all tests! ✓

Q3. Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num;++i)
    {
        if (num % i== 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 #include<stdio.h>
2
3 int count=0;
4 void Factor(int num){
5     for(int i=1; i<=num; ++i){
6         count++;
7         if(num%i==0){
8             count++;
9             //printf("%d ",i);
10        }
11        count++;
12    }
13    count++;
14    printf("%d",count);
15 }
16
17 int main(){
18     int n;
19     scanf("%d",&n);
20     Factor(n);
21 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Q4. Convert the following algorithm into a program and find its time

complexity using counter method.

void function(int n)

```
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 #include<stdio.h>
2
3 int count = 0;
4 void function(int n){
5     int c = 0;
6     count++;
7     for(int i=n/2; i<n; i++){
8         count++;
9         for(int j=1; j<n; j=2*j){
10             count++;
11             for(int k=1; k<n; k = k*2){
12                 count++;
13                 c++;
14                 count++;
15             }
16             count++;
17         }
18         count++;
19     }
20     count++;
21     printf("%d",count);
22 }
23
24 int main(){
25     int n;
26     scanf("%d",&n);
27     function(n);
28 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Q5. Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 #include<stdio.h>
2
3 int count = 0;
4 void reverse(int n){
5     int rev = 0, remainder;
6     count++;
7     while(n!=0){
8         count++;
9         remainder = n%10;
10        count++;
11        rev = rev * 10 + remainder;
12        count++;
13        n/=10;
14        count++;
15    }
16    count++;
17    count++;
18    printf("%d",count);
19 }
20
21 int main(){
22     int n;
23     scanf("%d",&n);
24     reverse(n);
25 }
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓